

Research paper

Does energy transformation improve green development efficiency? Evidence from Sichuan Province, China

Qiaochu Li^{a,*}, Peng Zhang^b, Junhua Chen^a

^a School of Economics and Management, Southwest Petroleum University, Chengdu 610500, China

^b School of Civil Engineering and Geomatics, Southwest Petroleum University, Chengdu 610500, China

ARTICLE INFO

Keywords:

Energy transformation

Green development efficiency

Network SBM model

Fixed effects regression model

ABSTRACT

Green sustainable development has become an important topic of concern to human society. Energy clean and low-carbon transformation is an important lever for promoting regional green development. Investigating the driving mechanisms of energy transformation is helpful to explore the effective path for accelerating the process of green development. Meanwhile, Sichuan Province is a demonstration province of clean energy and an important energy transportation hub in China, and it should play an exemplary role in the process of high-quality development. In this regard, this study uses the two-stage super-efficiency network SBM model to measure the green development efficiency based on the dual perspectives of green production and environmental governance. Based on historical data from 2006 to 2022, Technological innovation, industrial structure, and foreign direct investment are selected as transmission variables, and the fixed effects regression model is used to empirically analyse the direct and indirect impact effects of ET on GDE. The results show that the green development efficiency in Sichuan Province has gradually increased from 0.8303 in 2006–0.9270 in 2022, which is at a relatively high level. Energy transformation is helpful to improve the green development efficiency in Sichuan Province. However, due to insufficient complementary power sources and challenges in balancing energy supply and demand, the incentive level needs to be strengthened. Technological innovation, industrial structure, and foreign direct investment show indirect heterogeneous transmission effects in the process of influence.

1. Introduction

Green development is a new sustainable development model, which focuses on social, economic and environmental management to promotes human welfare through efficient utilization of natural resources (Fu and Geng, 2019). Global warming has promoted the clean and low-carbon transformation of energy supply and demand structure (Sun, 2022). In the process of accelerating sustainable development, human beings strive to reduce the utilization of fossil fuels and continuously promote energy transformation (ET) and upgrading. Building a low-carbon energy system by developing new energy and reducing traditional fossil energy consumption is an important measure to improve the green development efficiency (GDE) (Lau et al., 2023). Major countries around the world give priority to ET as the breakthrough point of green and low-carbon development and have formulated relevant policies (Battisti, 2023). EU member states have presented the REPowerEU plan to take actions to promote cheaper, safer and sustainable energy transition. The United States has issued Inflation

Reduction Act, and Japan has formulated The Energy and Environment Innovation Strategy for 2030. The report of the 20th National Congress of the Communist Party of China proposed to promote green development and harmonious coexistence between humans and nature, which is a new requirement and plan for the construction of ecological civilization. Promoting green transformation and ensuring energy security are inevitable policy orientations for achieving high-quality and sustainable development. Meanwhile, the implementation Plan for Promoting High-Quality Development of New Energy in the New Era, the 14th Five-Year Plan for Renewable Energy Development, and the 14th Five-Year Plan for Modern Energy System issued by China National Energy Administration all stress the need to drive green development through energy clean and low-carbon utilization.

As the largest developing country and the second largest population in the world, China's energy structure is highly dependent on fossil energy such as coal and oil; has a low utilization rate of clean energy; and can be characterized by "rich coal, less oil, gas" (Wan et al., 2023). Environmental pollution and ecological destruction caused by

* Corresponding author.

E-mail address: 252651625@qq.com (Q. Li).

<https://doi.org/10.1016/j.egy.2024.07.008>

Received 27 March 2024; Received in revised form 29 June 2024; Accepted 3 July 2024

Available online 6 July 2024

2352-4847/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

large-scale fossil energy consumption are becoming increasingly serious, and significantly restricts the green development process in China. In order to cope with environmental problems such as climate change and air pollution, various regions in China have begun to carry out ET such as coal to gas, coal to electricity, and oil to gas. However, the implementation of ET is not smooth, for example, loose coal combustion often occurs in the core coal-forbidden area. Whether ET can improve GDE has become a key issue of social concern. As an important energy supply base in China, the energy development and utilization in Sichuan Province has made great contributions to national economic and social development (Jiang et al., 2023). However, with the rapid progress of industrialization and urbanization, the excessive consumption of fossil energy has also caused serious environmental pollution. According to the 2022 Sichuan Statistical Yearbook, regional non-clean energy consumption accounted for 56.20 % of the total energy consumption. In this regard, actively promoting the clean energy substitution and the construction of low-carbon energy system is an effective measure to overcome the drawbacks of energy development and improve environmental quality in Sichuan Province. With the advantage of resource endowment, it is an effectual way to empower green development to cultivate natural gas and electricity as the main alternative energy sources (Wei et al., 2022).

In the international academic circle, the concept of green development has not been clearly put forward. The common terms include green economy, green growth, etc. Although the terms are different, they have the same theoretical core (Tang et al., 2023). It was first cited in the field of green economy by British scholar David Pearce in 1989, which emphasized the relationship between social development and ecological conditions (Jim, 2014). In further research, the connotation of green development has been extended from different perspectives, forming a new concept with the core of economic growth, environmental responsibility, and industrial transformation. On this basis, GDE evaluates the social, economic, and ecological coordination and comprehensive benefits (Haaland and van den Bosch, 2015; Olawumi and Chan, 2018) from the efficiency dimension. The concept of green development studied in this paper is mainly that the scale and speed of economic development are limited within the range of resource carrying capacity and environmental capacity, so as to maximize the resource utilization efficiency and minimize the environmental pollution, and finally achieve a coordinated state among economic development, resource utilization and environmental protection.

In this case, studying the impact of ET is of great significance for green, low-carbon, and high-quality development in China and even the whole world (Wojtkowska-Lodej and Jakubów, 2022). Recently, the government of China proposed to develop new quality productivity represented by technological innovation. During the process of ET, strengthening technological innovation is an important means to promote the high-quality and efficient utilization and further accelerate green development (Xu et al., 2023). The production and operation of the three major industries are important carriers of energy activities. The mutual transformation between different industries and the internal adjustment of each industry will have an important impact on the energy and environmental system (Ibrahim, 2022a). Meanwhile, under the influence of trade globalization, the flow of factors such as management, technology and talents driven by foreign investment activities significantly affects the energy utilization mode and the low-carbon development process (Houari, 2022). In this regard, different transmission variables play an important role in the influence process, and therefore conducting a comparative analysis of the direct and indirect impacts of ET on GDE can help to formulate differentiated driving paths and development strategies (Khan et al., 2022a). Considering the historical process of green development in China and the availability of statistical data, the period from 2006 to 2022 is selected as the research time range. It takes one year as a time unit and covers the whole period of 17 years. From this perspective, the major contributions of this study are as follows: (1) this article constructs the super-efficiency slacks-based

measure (SBM) model of two-stage hybrid network structure, and measures the GDE of Sichuan Province based on the dual perspectives of green production and environmental governance. Thus, the evaluation perspective of GDE can be expanded, and it can help solve the problem of distinguishing and sorting when multiple decision-making units (DMUs) are effective at the same time. (2) Policy and investment indicators such as the number of related laws and policies, the number of new projects and fixed assets investment quota are incorporated into the ET evaluation index system. Thus, it can make up for the deficiency that the existing researchs ignore the important impact of political and investment elements on the level of ET. (3) Technological innovation, industrial structure and foreign direct investment are selected as transmission variables, and the direct and indirect impact effects of ET on GDE are empirically discussed based on the fixed effects regression model, finally targeted policy implications are proposed.

The concept of green development is essentially to realize the rational material transformation between humans and nature. As an important carrier of material transformation, energy production and consumption play a key role in the process of green transformation development. Investigating the driving mechanisms of ET is helpful to explore the effective path for improving regional GDE. It is of great significance for the sustainable development of human society. Sichuan Province is a demonstration province of clean energy and an important energy transportation hub in China. According to the Statistical Yearbook of Sichuan Province, in 2022, the provincial natural gas output was $554.10 \times 10^8 \text{ m}^3$, and the hydropower export reached $1558.05 \times 10^8 \text{ kWh}$, both ranking among the top in the country. In this regard, Sichuan Province should play an exemplary role in the process of high-quality development, so as to contribute the "Sichuan Energy Plan" to the green development of the western region and even the whole country (Bamisile et al., 2022). Therefore, this study is meaningful to fully understand the GDE and the overall ET level of China's important clean energy producing areas. Meanwhile, by clarifying the heterogeneous influence mechanism of different paths under multi-agent game, we can provide goal orientation and path guidance for achieving high-quality green transformation of the whole society.

The rest of this article is structured as follows: The relevant literature review is in Section 2. Variable selection and data processing are in Section 3. Empirical analysis results are in Section 4. Main conclusions and policy implications are in Section 5. A two-step approach was used to investigate the influence mechanism of ET on GDE, as shown in Fig. 1.

2. Literature review

In recent years, improving the green consumption level of energy and accelerating the low-carbon transformation of energy system have become hot topics of social concern (Chen et al., 2024). From the perspective of concept universality, energy transformation is closely linked with the concept of green development (Zakari et al., 2022). The structure of literature review is shown in Table 1.

The rise of sustainable development theory promotes the development of green development evaluation models. There are mainly two ideas in the academic circle: (1) one is to evaluate the green development level according to the establishment of comprehensive indicators, including analytic hierarchy process (AHP), principal component analysis (PCA) and technique for order preference by similarity to an ideal solution (TOPSIS) methods. Liu (Liu, 2021) used the analytic hierarchy process (AHP) method to calculate the resource and environment cost accounting and analyze the green development level of different prefecture-level cities in the period of 2007–2017. Long et al (Long et al., 2021). constructed the evaluation index system for GDE from the four dimensions of economy, society, resources and environment, and selected the TOPSIS method to carry out empirical analysis on 30 coal resource-based cities in China. According to different research objects, the index system constructed by scholars is not the same, but it is nothing more than starting from the aspects of social construction,

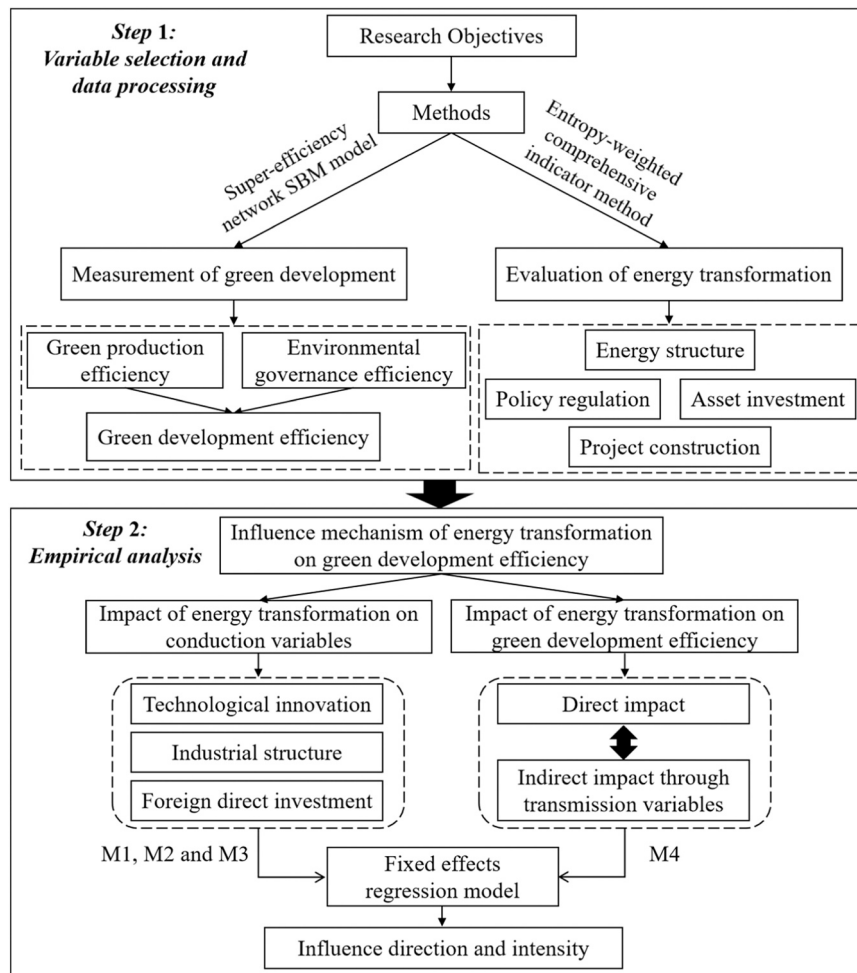


Fig. 1. The methodology framework.

economic level, environmental protection and resource utilization (Wang et al., 2018; Saša Stjepanović et al., 2017; Shi et al., 2016; Yue et al., 2016). (2) The second is to measure the green development from the perspective of efficiency by combining data envelopment analysis (DEA) and its extended model (SBM model, super efficiency model, etc.), which does not require pre-set production functions and indicator weights. Ge et al (Ge et al., 2023). used the single-layer SBM model to measure the agricultural GDE in China, and empirically tested the heterogeneous effects of registered residence urbanization and employment structure on GDE. Hu et al (Hu et al., 2023). measured the GDE of 41 prefecture-level cities in Yangtze River Delta region from 2000 to 2018 by using a non-radial super-efficiency DEA model, the results show that the GDE presented significant phase characteristics. Zhu et al (Zhu et al., 2019). and Lyu et al (Lyu et al., 2022). measured the GDE by using SBM model with undesirable outputs from the perspective of provinces and cities respectively, and evaluated the heterogeneity of spatial distribution.

As the ET cannot be directly observed, it needs to be evaluated using mathematical programming or econometrics models and theoretical tools. the main evaluation paradigm in academic circles is to first construct a comprehensive indicator system that represents energy structure characteristics such as the level of oil supply, and the proportion of new energy consumption; then, based on analytic network process (ANP), factor analysis, entropy weight and other model methods, the level of ET is evaluated (Wan, 2023). Khan et al (Khan et al., 2022b). established an indicator system of ET including multiple factors such as new energy generation installed capacity, consumption of fossil energy and imports of coal and oil, and carried out empirical

evaluations on Switzerland, Sweden and Denmark based on factor analysis respectively. Liang et al (Liang et al., 2024). also used this method to conduct empirical research on listed companies in the Chinese energy industry. Zhang et al (Yang and Wang, 2022). established the evaluation system of low-carbon transformation capability of urban regional comprehensive energy system based on the dual dimensions of low-carbon situation and market capacity, and carried out quantitative analysis by using ANP method. Zou et al (Zou et al., 2023). constructed an integrated economic-social-energy-environmental evaluation framework, and analyzed the transformation index of energy green consumption in eastern, central, western and northeastern regions of China from 2011 to 2016 based on entropy weight method, the results indicated that the ET index at the national level showed a trend of stagnation and decline, mainly because most regions of China were still constrained by the fossil fuel-dominated energy structure. Hou et al (Hou et al., 2021). established an indicator system of energy sustainable transformation from the dimensions of security, efficiency, cleanliness and low-carbon emissions, and conducted the comparative analysis of 30 provinces in China by the improved entropy weight-cloud model, the results showed that energy security and low-carbon emissions play a vital role in achieving sustainable transformation.

Nowadays, scholars have carried out research on the impact of factors such as mandatory environmental regulation (Zhan et al., 2023), economic complexity (Wang et al., 2023), urban development (Huang et al., 2023), digital economy (Tian and Zhou, 2022), manufacturing intelligence (Li and Ling, 2023), and smart transportation (Zhao and Dong, 2023) on GDE. However, the research on the relationship between ET and GDE needs to be further improved. Adam et al (Adam, 2022).

Table 1
Literature review structure.

Research object	Dimension	Summary	References
Measurement of green development	Comprehensive indicator measurement	Evaluating according to the establishment of comprehensive indicators, including analytic hierarchy process (AHP), principal component analysis (PCA) and technique for order preference by similarity to an ideal solution (TOPSIS) methods	(Bamisile et al., 2022; Chen et al., 2024; Zakari et al., 2022; Liu, 2021; Long et al., 2021; Wang et al., 2018)
	Efficiency measurement	Evaluating from the perspective of efficiency by combining data envelopment analysis (DEA) and its extended model (SBM model, super efficiency model, etc.)	(Saša Stjepanović et al., 2017; Shi et al., 2016; Yue et al., 2016; Ge et al., 2023)
Evaluation of ET	Comprehensive indicator measurement	Evaluating according to the establishment of comprehensive indicators, including analytic network process (ANP), factor analysis, entropy weight methods	(Hu et al., 2023; Zhu et al., 2019; Lyu et al., 2022; Wan, 2023; Khan et al., 2022b; Liang et al., 2024)
Impact of factors on GDE	GDE is influenced by diverse socio-economic factors	Influenced by mandatory environmental regulation, economic complexity, urban development, digital economy, manufacturing intelligence, smart transportation, etc.	(Yang and Wang, 2022; Zou et al., 2023; Hou et al., 2021; Zhan et al., 2023; Wang et al., 2023; Huang et al., 2023)
	Relationship between ET and GDE	Research perspective mainly focuses on the relationship between energy structure adjustment and GDE	(Tian and Zhou, 2022; Li and Ling, 2023; Zhao and Dong, 2023; Adam, 2022)

constructed the analysis framework of ET and GDE in the United Arab Emirates under the background of Three Seas Initiative, and the relationship between variables was determined by revealing the contribution of energy structure changes to green economy. However, the model relied on large amount of historical data to ensure the reliability of analysis results, so it had certain limitations. Hummera et al (Hummera et al., 2022). used autoregressive analysis method to investigate the influence mechanism of green economy, new energy and sustainable development in 12 Asian economies, and the relationship between ET and GDE under the assumption of environment Kuznets curve was analysed. The results showed that national policy makers should focus on building a new energy system in order to control energy-related environmental pollution more quickly. Yu et al (Yu et al., 2022). used the spatial econometric panel regression model to investigate the influence of biomass resource endowment on GDE, and the results showed that provincial biomass energy policy might not play its due role in promoting GDE because of the high cost of biomass energy. Zhang et al (Zhang et al., 2023). analyzed the impact of energy consumption structure change on green total factor productivity based on the spatial

econometric model, and the results indicated that the influence direction and degree showed regional heterogeneity characteristics due to multiple external factors such as economic development, resource endowment, industrial structure, technological innovation, etc.

So far, the research on the relationship between ET and GDE has made some progress in theory and practice, but the existing research has limitations in the following aspects: (1) the measurement of GDE only considers the green production process, and the important contribution of environmental governance process to efficiency improvement is ignored. (2) The research perspective mainly focuses on the relationship between energy structure adjustment and GDE. However, ET is a systematic project, and indicators such as policy regulation, project construction and asset investment cannot be ignored (Ibrahim, 2022b). (3) Most studies mainly discuss the direct impact of ET on GDE. Although the control variables have been considered, the indirect impact through key transmission variables such as technological innovation, industrial structure, and foreign direct investment is not taken into account.

3. Variable selection and data processing

3.1. Measurement of GDE

3.1.1. Network SBM model

DEA method originates from the concept of relative efficiency evaluation. Based on the premise that the input or output of DMUs remains unchanged, the difference between the production frontier and the optimal frontier of DMUs is calculated through reasonable mathematical programming, and it is used as a basis to measure whether DMUs are relatively effective (Yang et al., 2023). DEA method is suitable for solving the efficiency analysis problem of complex structures with multiple inputs and outputs.

The traditional DEA method treats the internal system as a "black box". The indicator variables first enter the input end, pass through the "black box" and finally output from the terminal, thus forming a basic single-layer structure. The internal transformation process within the system does not need to be investigated deeply, so the model operation is simple and convenient. However, with the continuous development of human civilization, the scientific problems to be solved tend to be diversified and complicated, and the applicability of traditional DEA method is decreasing. It is urgent to break through the internal of DMUs and decrypt the specific operation mechanism of the "black box" to fundamentally solve the problem of low efficiency. Combined with the slacks-based measure method, Tone et al (Tone and Tsutsui, 2009). first proposed a network SBM model, which extended the application boundary of DEA method to the research scenarios with free changes in input and output variables. When the number of input and output indicators is large, the number of effective DMUs will also be large (Lu and Zhang, 2023). Meanwhile, when investigating the time series data of the same research object, the maximum values obtained by the DEA model are fixed at 1, and the efficiency differences of different effective DMUs cannot be further distinguished. In this regard, the network SBM model does not consider the super-efficiency setting, so it is difficult to rank the efficiency values of multiple DMUs when they are all in 100 % effective state, which is not conducive to further data regression analysis. At the same time, the measurement of GDE cannot ignore the impact of unexpected output such as industrial wastes. Therefore, this article uses the super-efficiency network SBM considering unexpected output to quantify the GDE. Each DMU is assumed to be composed of k subsystems ($k = 1, 2, \dots, K$), and the model equation is as follows.

$$\rho_0^* = \min \frac{\sum_{k=1}^K w^k \left[1 + \frac{1}{m_k} \left(\sum_{i=1}^{m_k} \frac{s_{io}^{k-}}{x_{io}^k} \right) \right]}{\sum_{k=1}^K w^k \left[1 - \frac{1}{n_{1k} + n_{2k}} \left(\sum_{i=1}^{n_k} \frac{s_{io}^{gk}}{y_{io}^{gk}} + \sum_{i=1}^{n_k} \frac{s_{io}^{bk}}{y_{io}^{bk}} \right) \right]}$$

$$\text{s.t.} \left\{ \begin{array}{l} \sum_{i=1}^k w^i = 1, w^i \geq 0 \\ x_0^k \geq \sum_{j=1}^n x_j^k \lambda_j^k + s_0^{k-} \\ y_0^{gk} \leq \sum_{j=1}^n y_j^{gk} \lambda_j^k + s_0^{gk} \\ y_0^{bk} \geq \sum_{j=1}^n y_j^{bk} \lambda_j^k - s_0^{bk} \\ z_0^{(k,s)} \lambda^s = z^{(k,s)} \lambda^k \\ \varepsilon \leq 1 - \frac{1}{n_{1k} + n_{2k}} \left(\sum_{i=1}^{n_k} \frac{s_{io}^{gk}}{y_{io}^{gk}} + \sum_{i=1}^{n_k} \frac{s_{io}^{bk}}{y_{io}^{bk}} \right) \\ s_0^{k-}, s_0^{gk}, s_0^{bk}, \lambda^k \geq 0 \end{array} \right. \quad (1)$$

In this equation, ρ_0^* is the efficiency value of DMUs; w^k is the weight value, which indicates the relative importance of the I -th subsystem of DMUs; x_{io}^k , y_{io}^{gk} and y_{io}^{bk} are the input, expected output and unexpected output vectors of the I -th subsystem of DMUs, respectively; s_0^{k-} , s_0^{gk} and s_0^{bk} are relaxation variables; m_k and n_k are the input and output quantities of the I -th subsystem of DMUs, respectively; λ^k is the intensity vector; $z_0^{(k,s)}$ is the linked variable, which simultaneously represents the output variable of the k -th subsystem and the input variable of the s -th subsystem.

3.1.2. Frame structure and indicator selection

Energy utilization not only promotes economic development, but also brings severe challenges to the ecological environment. Therefore, it is of great significance to synergistically strengthen the high-quality development of regional economy and the high-level protection of ecological environment. The main process of green development can be divided into green production subsystem and environmental governance subsystem, so as to effectively deconstruct complex social operation problems. For the green production subsystem, the labor, capital and energy resources required for production are transformed into corresponding economic benefits through complex transformation process, and environmental pollutants are produced at the same time. For the environmental control subsystem, effective removal of pollutants is finally realized by investing sufficient pollution control facilities and operating expenses. Meanwhile, the environmental pollutant is selected as the link index of the two subsystems. The model structure framework is shown in Fig. 2, and the measurement indicators are summarized in Table 2.

3.1.3. Calculation of GDE

The green production efficiency, environmental governance efficiency and GDE of Sichuan Province from 2006 to 2022 are calculated based on network SBM model, and the results for the whole period of 17 years are shown in Fig. 3.

According to Fig. 3, the GDE of Sichuan Province during the period of 2006–2022 showed a fluctuating upward trend. This was mainly due to the fact that regional development adhered to the goal of building a national clean energy demonstration province, and the leverage effect of green taxation and the incentive effect of governance investment were

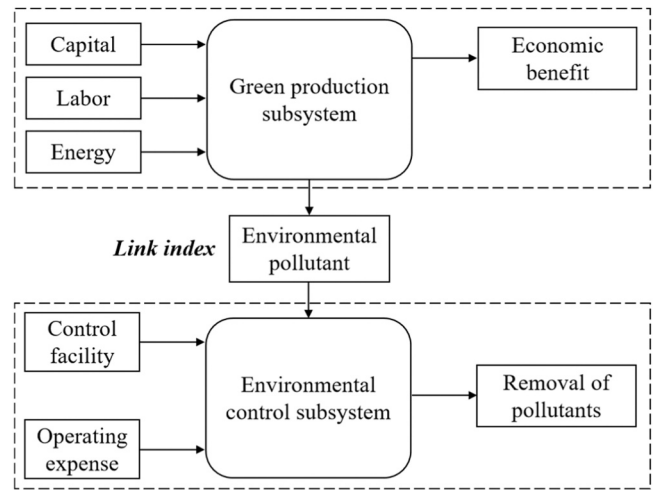


Fig. 2. Model structure framework.

taken into account. At the same time, regional construction was supported by the development of central cities and the growth of advantageous industries, while vigorously promoting the construction of socialist ecological civilization.

During the 11th Five-Year Plan period (2006–2010), the GDE of Sichuan Province showed a downward trend on the whole. This was mainly because during this period, China's economic and social development had achieved important growth, and its fiscal revenue and export revenue reached a record high. The green production process was in its early stage. It is worth noting that since 2008, under the dual impact of the international financial crisis and major natural disasters such as the Wenchuan earthquake, the focus of regional development had been on the expansion of domestic demand and post-disaster reconstruction, and environmental governance had given way to the development of infrastructure and basic industries, and its efficiency showed a significant downward trend. However, the 17th National Congress of the Communist Party of China in 2007 included the construction of ecological civilization in the report for the first time. Meanwhile, In 2010, the 5th Plenary Session of the 17th Central Committee of the Communist Party of China put forward that the construction of a resource-saving and environment-friendly society should be an important focus to accelerate the transformation of economic development mode, which laid a solid foundation for the improvement of GDE in Sichuan Province in the future.

During the 12th Five-Year Plan period (2011–2015), the GDE of Sichuan Province showed a steady upward trend. This was mainly because the 12th Five-Year Plan was the first green development plan and an important starting point for green modernization in China. Especially since the 18th National Congress of the Communist Party of China, China had unswervingly taken the road of ecological priority and green development, and had achieved remarkable green development achievements during this period. Both green production efficiency and environmental governance efficiency showed an obvious upward trend. However, at the same time, the pollution of fog and haze continued to spread and intensify in 2013, and different degrees of air pollution appeared in various regions of Sichuan Province. Moreover, with the investment quota of 1489.26 million yuan, the regional waste gas treatment capacity was only 488.79 million m³/h. The environmental governance efficiency was low, which led to the reduction of GDE from 0.3634 in 2012–0.2696 in 2013. Thus, the problem of clean energy utilization, represented by coal smoke pollution, became increasingly prominent. The energy structure optimization such as coal to gas and oil to gas needed to be carried out urgently, while the effectiveness of air pollutant treatment needed to be enhanced.

During the 13th Five-Year Plan period (2016–2020), the GDE of

Table 2
Measurement indicators of GDE.

Subsystem type	Indicator type	Specific indicator	Measurement method	Basic data source
Green production subsystem	Input index	Labor input	Number of regional employees	Sichuan Statistical Yearbook
		Capital input	Regional capital stock	Sichuan Statistical Yearbook
		Energy input	Total regional energy consumption	China Energy Statistical Yearbook
Link index	Output index	Economic benefit output	Regional Gross domestic product (GDP)	Sichuan Statistical Yearbook
		Environmental pollutant	Total amount of the three wastes (waste gas, waste water and industrial residue)	China Environmental Yearbook
Environmental control subsystem	Input index	Facility input	Number of the three wastes treatment facilities	China Environmental Statistical Yearbook
		Expense input	Completed investment quota for the three wastes treatment project	China Environmental Yearbook; Sichuan Environmental Statistics Bulletin
	Output index	Governance effectiveness	The three wastes treatment capacity	China Environmental Yearbook

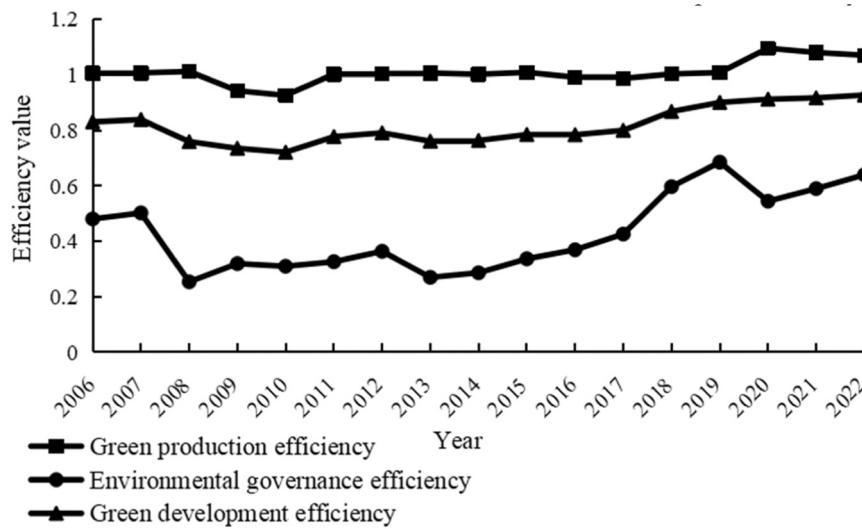


Fig. 3. Evolution trend of efficiency values in Sichuan Province from 2006 to 2022.

Sichuan Province improved rapidly. This was mainly because the concept of green development was included in the 13th Five-Year Plan for the first time. During this period, major ecological protection and restoration projects were further promoted, which led to an increase in the environmental governance efficiency from 0.3364 in 2015–0.6849 in 2019 (Since 2020, environmental governance had been affected by the black swan event such as COVID-19 to some extent). The environmental governance efficiency showed a linear growth trend, and the green production efficiency had also increased from 0.9915 in 2016–1.0961 in 2020. In this regard, green development became a new driving force for high-quality economic growth. The green production efficiency declined briefly in the early period of the 13th Five-Year Plan, and then began to increase year by year. This was mainly because the total energy consumption in Sichuan Province increased significantly in 2016 and 2017, up by 2.45 % and 5.04 % respectively compared with 2015, which reversed the downward trend of total energy consumption since 2013. However, the GDP growth rate did not improved significantly at the same time, so the problem of low energy utilization efficiency had affected the improvement of regional GDE.

During the early period of the 14th Five-Year Plan (2021–2022), with the precise control of the adverse effects of the COVID-19 epidemic, social operation gradually entered a new mode in the post-epidemic era. Green production had to give way to the national economic recovery, and its efficiency had decreased to 1.0706 in 2021. However, with the implementation of "carbon peaking and carbon neutrality", environmental governance work had been gradually strengthened under policy incentives, and its efficiency had increased from 0.9121 in 2020–0.9270

in 2022. Therefore, although the growth rate slowed down as a whole, the GDE still maintained an upward trend during this period. In summary, optimizing energy structure and improving energy utilization efficiency are the two major focuses of ET, and also the key to promote the GDE of Sichuan, the western region and even China.

3.2. Evaluation of ET level

3.2.1. Establishment of evaluation indicator system

The development and utilization of coal resources in Sichuan Province does not match the resource endowment, and the demand basically depends on the energy allocation from outside the province. There are significant hidden dangers of energy overexploitation, and it is urgent to carry out clean substitution in the process of energy transformation. At the same time, natural gas and hydropower resources in Sichuan Province have unique endowment advantages and great development potential, so that the regional ET should focus on the substitution of natural gas and electric energy for coal resources.

Considering the principles of scientificity, comprehensiveness, target and availability, the following indicators are selected: (1) The proportion of natural gas and electric energy consumption to total energy consumption (X1), which reflects the clean state of energy structure in Sichuan Province. The greater the proportion, the higher the energy clean transformation level. (2) The amount of coal transferred from other provinces (X2), which reflects the dependence degree of Sichuan Province on coal resources. The less the transfer volume, the stronger the implementation of ET. (3) Number of laws and policies related to ET

issued within one year (X3), which reflects the intensity of ET measures carried out by government departments in Sichuan Province. The more relevant laws and policies are issued, the more effective the ET measures under the control of the government's "tangible hand". (4) Numbers of new projects (X4) and fixed assets investment quota (X5) in natural gas and electric energy related industries, which reflects the investment in ET in Sichuan Province. The more projects and funds are invested, the more favorable it is to promote the optimization of regional energy structure and the improvement of clean energy utilization level. The basic data of the above indicators come from China Statistical Yearbook, China Energy Statistical Yearbook, China Environmental Statistical Yearbook and Sichuan Statistical Yearbook from 2007 to 2023.

3.2.2. Measurement of ET level

Due to the selection of multiple indicators, direct regression analysis may lose a certain level of freedom. Therefore, the entropy weight method can be used to synthesize the multiple indicators involved in ET into a comprehensive level index. The entropy weight method is a mathematical analysis method of objective assignment, which is used to determine the dispersion degree of indicators. Entropy is a measure of uncertainty. If the differences among the elements in a certain system are small and the elements are unified and orderly, it indicates that the system has a large amount of effective information and a small entropy value. Thus, it will be given a small weight, otherwise will be given a large weight. In this regard, the results are objective and reasonable through scientific mathematical analysis. Because the entropy weight method uses the inherent information of evaluation indicators to determine their utility value, it can avoid the deviation caused by subjective factors to some extent. The weight value given by the entropy weight method is more reliable than subjective weighting methods such as Delphi model and AHP. Based on the entropy weight method, the level score of ET can be obtained by multiplying the basic data of relevant indicators in Sichuan Province from 2006 to 2022 with the corresponding weight values. Based on this solution, the obtained weight coefficients of evaluation indicators are summarized in Table 3.

3.3. Selection of variables for econometric regression analysis

In order to investigate the influence mechanism of ET on GDE, taking into account the conduction effects of technological innovation, industrial structure and foreign direct investment, as well as the impact of control variables, the following variables are ultimately selected: (1) Green development efficiency (GDE), which is represented by the calculated efficiency value based on the super-efficiency SBM model of two-stage hybrid network structure, which will not be repeated here. (2) Energy transformation (ET), which is represented by the calculated level index based on the entropy-weighted comprehensive indicator evaluation method. (3) Technological innovation (TI), which is represented by the proportion of science and technology expenditure in the total expenditure of general public budget in Sichuan Province. The more funds are invested in scientific research, the higher the attention is paid to technological innovation. Thus, the more likely it is to provide advanced scientific achievements for enterprises and other economic entities, and the more likely it is to promote the improvement of production technology, technological process and product quality. (4) Industrial structure (IS), which is represented by the proportion of gross industrial production to regional GDP. Industry plays an important role in the process of regional economic and social development, but it consumes large amount of fossil energy in daily production operation. The adjustment of industrial internal structure and the change of its

scale will both affect the GDE (Xu and Zhou, 2023). (5) Foreign direct investment (FDI), which is represented by the amount of regional foreign direct investment. With the continuous improvement of Sichuan Province's economic opening to the outside world and the continuous expansion of investment attraction scale, the scale of foreign direct investment has increased substantially, which further affects the regional GDE by changing the industrial development mode. (6) Economic development (ED), which is represented by the regional per capita GDP. Regional economic development level will significantly affect socio-economic factors such as technological innovation level, industrial structure characteristics, and foreign direct investment scale, so it is regarded as a control variable. (7) Human capital (HC), which is represented by the per capita education years of the population aged 6 and above in Sichuan Province. Human capital level represents the ability of regional labor force to create wealth as an economic subject, which is manifested as the sum of production knowledge, labor and management skills, and health quality contained in human beings (Shen et al., 2023). Human capital level will greatly affect the regional industrial structure characteristics, so it is regarded as a control variable. The basic data of the above variables come from China Statistical Yearbook, Sichuan Statistical Yearbook, and China Population and Employment Statistical Yearbook, and it covers the whole period of 17 years from 2006 to 2022.

4. Empirical analysis

4.1. Model setting

Before clarifying the influence mechanism of ET on GDE, it is necessary to first investigate the relationship between independent variables and transmission variables. ET is taken as the key explanatory variable, and technological innovation (TI), industrial structure (IS) and foreign direct investment (FDI) are taken as the explained variables respectively. The historical data of 17 years from 2006 to 2022 are used for regression analysis to explore the influence effects of factors. In this regard, the models M1 to M3 are established as follows.

$$M1: TI_t = \alpha_1 + \beta_{11} \ln ET_t + \beta_{12} \ln ED_t + \beta_{13} \ln IS_t + \varepsilon_t \quad (2)$$

$$M2: IS_t = \alpha_2 + \beta_{21} \ln ET_t + \beta_{22} \ln ED_t + \beta_{23} \ln HC_t + \varepsilon_t \quad (3)$$

$$M3: FDI_t = \alpha_3 + \beta_{31} \ln ET_t + \beta_{32} \ln ED_t + \beta_{33} \ln IS_t + \varepsilon_t \quad (4)$$

In this equation, t is the year; α and β are parameters to be estimated; ε_t is the random error term.

In order to deeply investigate the impact of ET on GDE, GDE is selected as the explained variable, and ET and transmission variables are selected as the main explanatory variables. The model M4 is established as follows:

$$M4: GDE_t = \alpha + \beta_1 \ln ET_t + \beta_2 \ln TI_t + \beta_3 \ln IS_t + \beta_4 \ln FDI_t + \varepsilon_t \quad (5)$$

4.2. Empirical analysis results

Combined with models M1 to M4, data regression processing is carried out based on the fixed effects regression model, and the specific results are summarized in Table 4. Meanwhile, the direct impact and

Table 3
The weight coefficients of evaluation indicators.

Evaluation indicator	X1	X2	X3	X4	X5
Weight coefficient	0.2278	0.1797	0.2967	0.1649	0.1309

Table 4
Regression model estimation results.

Variable	M1	M2	M3	M4
Constant term	-17.6063**	-1.8200*	-11.0770*	0.9530*
ET	0.6394	0.0500*	-1.7093*	0.1451**
TI	-	-	-	-0.0400**
IS	-3.2922*	-	1.0236**	-0.2706*
FDI	-	-	-	0.0503*
ED	3.3835***	0.7329***	0.5775***	-
HC	-	-0.4083*	-	-

indirect transmission effects of ET on GDE is shown in Fig. 4. Referring to the research of Dagar et al (Dagar and Malik, 2023), ***, ** and * represent significance level at 1 %, 5 % and 10 % respectively. + and - represent positive motivation effect and negative inhibition effect respectively.

4.2.1. Impact of ET on transmission variables

4.2.1.1. Analysis on the direct impact of ET on technological innovation.

According to the regression results of model M1, although the regression coefficient reached 0.6394, it was not significant within 10 % level. The possible reason for this phenomenon is that ET can force enterprises to switch to clean energy utilization in their daily production operations through policy regulatory measures such as carbon tax, energy tax and environmental tax. At this time, enterprises may carry out technological research and innovation, optimize and upgrade products and process flows to match the clean energy structure, so the regression coefficient is positive. But at the same time, ET will significantly increase the energy costs and investment quota in new equipments for enterprises, thus squeezing the original scientific research and development funds (Liu et al., 2023). In addition, traditional fossil energy sources such as coal and oil products have the advantages of stability and high calorific value in industrial production. Meanwhile, the breakthrough technologies of efficient utilization of clean energy such as natural gas and new energy are still facing certain bottlenecks, so it is difficult for energy consuming enterprises to obtain significant economic benefits in the short term through the research and application of high-grade, precision and advanced technologies (Zhao et al., 2021). Therefore, enterprises may choose to ease the cost pressure by simplifying the pollution treatment process. In order to improve the significance of the driving effect of ET on technological innovation, it is urgent to enhance the compensation effect of technological innovation in the field of ET.

4.2.1.2. Analysis on the direct impact of ET on industrial structure. According to the regression results of model M2, with the improvement of ET level, the proportion of gross industrial production in Sichuan's regional GDP shows a certain upward trend. For every 1 % increase in ET level, the proportion of gross industrial production will correspondingly increase by 0.0500 %. Industry has always been an important pillar of economic and social development in Sichuan Province, and industrial production mainly depends on coal resources for energy supply. However, from the perspective of energy reserves in Sichuan Province, the proportion of coal is less than a quarter, so it basically relies on the transfer from other provinces to maintain regional energy demand. Resource endowment has become a barrier to the further development of provincial industry. On the one hand, with the

improvement of ET level, in order to adapt to the new energy consumption structure, enterprises will update and upgrade their own production processes and product categories (Child et al., 2018). Meanwhile, enterprises will increase the utilization of clean energy or change from resource-intensive industries to technology-intensive industries, and break through the barriers that restrict further development, thus promoting the high-quality increase of gross industrial production in Sichuan Province. On the other hand, with the continuous development of social progress and human civilization, human beings gradually realize the importance of green development. While pursuing the maximization of economic benefits, human beings have also begun to focus on the high-level protection of the ecological environment and seek the clean and low-carbon transformation of energy structure. However, the large amount of manpower, material resources, and financial resources required become a major obstacle. In the process of ET, the government will give preferential policies and financial support to enterprises that actively carry out energy structure optimization or use clean energy for production operations. In this regard, it can solve the cost obstacle of ET to a certain extent, and new development opportunities will be created for the whole industry. Finally, it can promote the sustainable development of industry in Sichuan Province while optimizing the internal structure.

4.2.1.3. Analysis on the direct impact of ET on foreign direct investment.

According to the regression results of model M3, with the improvement of ET level, the scale of foreign direct investment in Sichuan Province shows a certain downward trend. For every 1 % increase in ET level, the scale of foreign direct investment will correspondingly reduce by 1.7093 %. ET measures are closely linked to the production costs of energy consuming enterprises. With the increasing implementation of ET measures, the energy costs of enterprises have increased significantly, and gradually approached the external transfer cost (Yu et al., 2021). In order to obtain more economic profits, multinational enterprises that rely heavily on non-clean energy will tend to change their business policies and focus on investing in underdeveloped areas with low implementation of ET measures, thus causing the problem of pollution havens, which is consistent with the findings of Dagar et al (Dagar et al., 2022). As a national clean energy demonstration province, Sichuan Province has actively carried out ET based on the advantages of resource endowment in recent years, resulting in strict environmental regulations for foreign investment. Foreign investors can no longer obtain high profits by investing in industries with high pollution and high energy consumption, which leads to a continuous decline in the amount of regional foreign direct investment from \$8963.75 million in 2018 to \$3530.59 million in 2022. Overall, energy clean and low-carbon transformation will restrain the profit-oriented foreign investment

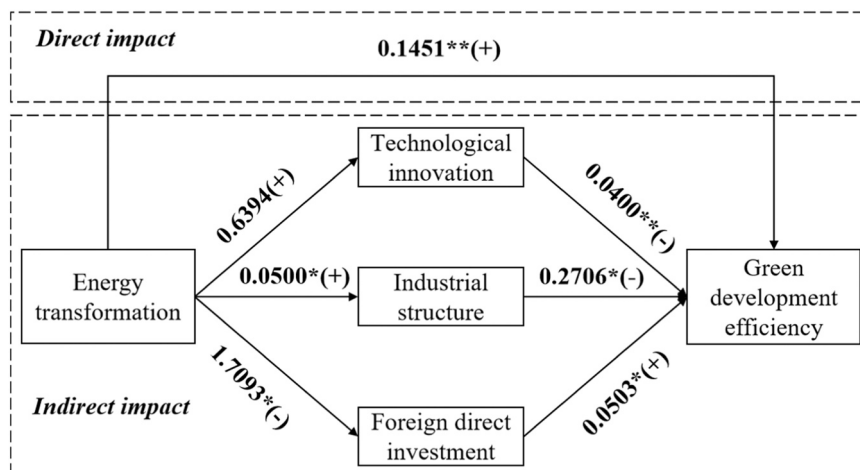


Fig. 4. The direct impact and indirect transmission effects.

enthusiasm to a certain extent. Under the new pattern of green rules of international trade, Sichuan Province should take advantage of the opportunity to accelerate the development of green foreign trade, and pay equal attention to improving the foreign investment scale and optimizing the foreign investment structure.

4.2.2. Impact of ET on GDE

4.2.2.1. Analysis on the direct impact of ET on GDE. According to the regression results of model M4, there is a positive correlation between ET and GDE in Sichuan Province. For every 1 % increase in ET level, the GDE will correspondingly increase by 0.1451 %. This indicates that during the research period, due to the improvement of ET level, the consumption proportion of clean energy such as natural gas and new energy in Sichuan Province has increased accordingly. Clean energy has the characteristics of low carbon and high efficiency characteristics of the clean energy, which helps enterprises to carry out green production (Guan and Barker, 2012). Meanwhile, the transformation of consumer orientation under policy guidance also helps to promote the launch of new green products, which is conducive to significantly reducing pollutant emissions and accelerating the improvement of regional GDE. However, at present, the incentive effect of ET on GDE in Sichuan Province still needs to be strengthened. The main reasons for this phenomenon are: on the one hand, electricity is not a primary energy source, and its production structure also includes a certain proportion of thermal power, which is supplied by burning non-clean energy such as coal. During the production process, some exhaust emissions such as sulfur dioxide, nitrogen oxides, and dust will still be discharged. At the same time, the installed capacity of hydropower in Sichuan Province is as high as 78 % (Guo et al., 2023). Due to the defects of regional distribution and seasonal fluctuation, it is difficult for thermal power to make up the power gap caused by insufficient hydropower output under special circumstances such as extreme high temperature weather, which is not conducive to the stable drive of ET to regional GDE. Therefore, in the future, Sichuan Province should strengthen power supply construction and realize multi-energy complementation, mutual assistance of hydropower and thermal power, synchronization of wind and solar energy, and give consideration to total quantity improvement and structural adjustment. On the other hand, due to the task of West-to-East Gas Transmission, the overall supply and demand situation of natural gas in Sichuan Province has gradually developed from self-production and self-use to both self-use and export. Under the impact of extreme weather and unexpected events, the balance of energy supply and demand will face significant challenges, which is also not conducive to the steady progress of regional green development. Therefore, in the future, gas storage facilities should be ensured to be fully stored while guiding production enterprises to increase and stabilize production in Sichuan Province (Li et al., 2023a).

4.2.2.2. Analysis on the indirect impact of ET on GDE through transmission variables. For the transmission effect of technological innovation, there is a negative correlation between technological innovation and GDE in Sichuan Province. For every 1 % increase in technological innovation level, the GDE will correspondingly decrease by 0.0400 %. Based on the analysis above, the ET in Sichuan Province has not played a significant role in promoting technological innovation. The possible reason for this phenomenon is that the implementation of ET increases the pollution control costs of related industries and squeezes economic profits, thus reducing the motivation and funding for local enterprises to carry out green technology innovation, which is consistent with the findings of Li et al (Li et al., 2023b). As a result, although some funds have been invested in scientific research and experimental development at the current stage, advanced technological achievements are mainly used to expand industrial production scale and update product performance, rather than to improve pollutant treatment capacity and reduce energy

consumption in production process (Zhou and Fan, 2023). Specifically, according to Sichuan Statistical Yearbook in 2022, there were 2338 application projects of scientific and technological achievements in Sichuan Province, including 558 projects related to industrial production, accounting for 23.87 %. However, there were only 60 projects related to environmental protection, accounting for 2.57 %. Therefore, the driving effect of ET on GDE in Sichuan Province through technological innovation needs to be strengthened.

For the transmission effect of industrial structure, there is a negative correlation between industrial structure and GDE in Sichuan Province. For every 1 % increase in the proportion of gross industrial production to regional GDP, the GDE will correspondingly decrease by 0.2706 %. Based on the analysis above, ET can help Sichuan Province break through the obstacles of further industrial development through energy consumption structure constraints and policy financial support, which agrees with the findings of Li et al (Li et al., 2023c). But on the whole, industry is an important pillar of economic and social development in Sichuan Province, and its energy utilization is highly intensive. In particular, industries such as steel, non-ferrous metals and petrochemicals, which are highly dependent on non-clean energy, occupy a high proportion in the region. As a result, the total energy consumption is large and its contribution to environmental pollution is high, which finally hinders the regional green development process.

For the transmission effect of foreign direct investment, there is a positive correlation between foreign direct investment and GDE in Sichuan Province. For every 1 % increase in the scale of foreign direct investment, the GDE will correspondingly increase by 0.0503 %. Based on the analysis above, in recent years, the ET measures in Sichuan Province has restricted foreign investment in industries with high pollution and high energy consumption, which is similar to the findings of Udemba et al (Udemba et al., 2023). on Norway. But at the same time, under the new pattern of green and sustainable development, foreign investment in environment-friendly industries will get new preferential policies, and it will help stimulate the formation of innovative investment effects under the pressure of cost minimization. On the one hand, foreign-funded enterprises attach importance to energy security and environmental protection. They apply advanced scientific research achievements to the production of green products and invest more money to carry out technological upgrading. As a result, the resource consumption and pollutant emission during the production process will be significantly reduced. Meanwhile, the collaborative governance of pollution reduction and carbon reduction will be strengthened, and it will finally effectively accelerate the regional green development process. On the other hand, local enterprises can also seize the opportunity of foreign investment to learn from international advanced technological achievements (Tariq et al., 2022). Thus, it is helpful to alleviate the problem of technological limitation and talent shortage, enhance the benefits of technological innovation in the field of environmental protection, and finally improve the regional GDE.

5. Conclusions and policy implications

This study uses the two-stage super-efficiency network SBM model to measure the GDE in Sichuan Province from 2006 to 2022 based on the dual perspectives of green production and environmental governance. Meanwhile, the comprehensive indicator measurement method is used to evaluate the level of ET. Further, technological innovation, industrial structure, and foreign direct investment are selected as transmission variables, and the direct and indirect impact effects of ET on GDE is empirically analysed. The main research conclusions are as follows: (1) The GDE in Sichuan Province has gradually increased from 0.8303 in 2006–0.9270 in 2022, which is at a relatively high level. (2) ET is helpful to improve the GDE in Sichuan Province. However, due to insufficient complementary power sources and challenges in balancing energy supply and demand, the regression coefficient is only 0.1451, so the incentive level needs to be strengthened. (3) Enterprises can alleviate

the cost pressure caused by ET through technological innovation, but it may lead to insufficient research funds and simplified pollution reduction processes. (4) ET can help the industrial sectors break through the obstacles of green and sustainable development through energy consumption structure constraints and policy financial support. (5) ET will limit foreign direct investment in industries with high pollution and high energy consumption, and improve regional GDE through innovative investment effect.

Based on the research findings, the following policy implications are proposed: (1) Following the overall approach of reducing coal, controlling oil, increasing electricity, stabilizing gas, and promoting new energy, it is committed to doing a good job in reducing fossil energy and adding renewable energy. Energy projects such as West-to-East Gas Transmission and West-to-East Power Transmission should be actively carried out, so as to coordinate energy supply security and low-carbon energy development. (2) It is urgent to promote the high-end industrial structure and actively accelerate industrial transformation and upgrading, as well as low-carbon development. The frontier industries of "new materials + new energy + ecological protection" such as key materials of lithium battery, hydrogen energy, high-performance engineering plastics and comprehensive utilization of industrial waste should be integrated. Regional industry should take the path of clean development, and the safe promotion of diversified, green, and stable energy supply need to be ensured. Specifically, industrial enterprises using natural gas as raw material or fuel should be encouraged to settle down, and the raw material advantages of the largest integrated petrochemical and refining project in the Southwest China should be fully utilized to supply aviation kerosene, high-quality gasoline and diesel, and basic chemical raw materials for the Chengdu-Chongqing area. (3) Research funding and talent investment should be increased, and the roadmap for tackling key problems in green and low-carbon technologies needs to be formulated. Relying on innovative platforms such as universities and research institutes, bottleneck technologies in the fields of clean coal utilization, unconventional natural gas exploitation, grid-connected and decentralized operation of renewable energy power generation should be cracked, and technological breakthroughs such as carbon capture, carbon sequestration and pollutant treatment should be carried out simultaneously. (4) The utilization efficiency of foreign direct investment should be given more attention, especially the transmission role of green technology progress. The strategy of investment promotion and capital selection needs to be formulated according to local conditions, and the scale and quality of foreign direct investment should be taken into account at the same time. The capital, competition, and technology spillover effects of foreign direct investment should be fully utilized to promote innovation. Meanwhile, it is necessary to pay attention to and avoid the problem of pollution havens.

This study offers some innovative insights, but it also has significant limitations that may open up new areas for further study. First, limited by the availability of data, the indicator system we built in this study, although objectively reflecting the level of ET, can further be optimized. Moreover, future research can consider combining the DEA theory with spatial econometric analysis model to more comprehensively investigate the influence mode and regional heterogeneity difference of ET on GDE.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was funded by the National Social Science Fund of China (Grant numbers 22&ZD105).

CRediT authorship contribution statement

Qiaochu Li: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Peng Zhang:** Writing – review & editing, Software, Resources, Investigation. **Junhua**

Chen: Writing – review & editing, Software, Investigation.

Declaration of Competing Interest

On behalf of all authors, I declare that I have disclosed all competing interests related to this work. If any exist, they have been included in the "declaration of interests" section of the manuscript.

Data availability

Data will be made available on request.

References

- Adam, K., 2022. Energy Transformation and the UAE green economy: trade exchange and relations with Three Seas initiative countries. *Energies* 15, 8410.
- Bamisile, O., Wang, X.K., Adun, H., Joseph Ejiyi, C., Obiora, S., Huang, Q., Hu, W.H., 2022. A 2030 and 2050 feasible/sustainable decarbonization perusal for China's Sichuan Province: a deep carbon neutrality analysis and EnergyPLAN. *Eng. Convers. Manag.* 261, 115605.
- Battisti, L., 2023. Energy, power, and greenhouse gas emissions for future transition scenarios. *Eng. Policy* 179, 113626.
- Chen, J.H., Li, Q.C., Zhang, P., Wang, X.Y., 2024. Does technological innovation efficiency improve the growth of new energy enterprises? Evidence from listed companies in China. *Sustainability* 16 (4), 1573.
- Child, M., Koskinen, O., Linnanen, L., Breyer, C., 2018. Sustainability guardrails for energy scenarios of the global energy transition. *Renew. Sust. Energ. Rev.* 91, 321–334.
- Dagar, V., Ahmed, F., Waheed, F., Bojnec, S., Khan, M.K., Shaikh, S., 2022. Testing the pollution haven hypothesis with the role of foreign direct investments and total energy consumption. *Energies* 15 (11), 4046.
- Dagar, V., Malik, S., 2023. Nexus between macroeconomic uncertainty, oil prices, and exports: evidence from quantile-on-quantile regression approach. *Environ. Sci. Pollut. R.* 30 (16), 48363–48374.
- Fu, J., Geng, Y., 2019. Public Participation, regulatory compliance and green development in China based on provincial panel data. *J. Clean. Prod.* 230, 1344–1353.
- Ge, P.F., Liu, T., Wu, X.X., Huang, X.L., 2023. Heterogenous urbanization and agricultural green development efficiency: evidence from China. *Sustainability* 15, 5682.
- Guan, D.B., Barker, T., 2012. Low-carbon development in the least developed region: a case study of Guangyuan, Sichuan province, southwest China. *Nat. Hazards* 63 (3), 243–254.
- Guo, X.R., Wang, H.M., Xiao, Y., Liu, G., Wang, W.Q., 2023. Reassessing the cost of hydropower for multi-function hydropower bases: a case study from Sichuan. *J. Clean. Prod.* 413, 137422.
- Haaland, C., van den Bosch, C., 2015. Challenges and strategies for urban green-space planning in cities undergoing densification: a review. *Urban. Urban Green.* 14 (4), 760–771.
- Hou, X.R., Lv, T., Xu, J., Deng, X., Liu, F., Pi, D.Y., 2021. Energy sustainability evaluation of 30 provinces in China using the improved entropy weight-cloud model. *Ecol. Indic.* 126, 107657.
- Houari, O., 2022. Uncertainty shocks and business cycles in the US: new insights from the last three decades. *Econ. Model.* 109, 105762.
- Hu, M.J., Li, Z.J., Hou, B., 2023. The influencing effect of tourism economy on green development efficiency in the Yangtze River Delta. *Int. J. Environ. Res. Public Health* 20, 1072.
- Huang, J., Shanguan, Z.H., Gui, T.T., Liu, J.P., 2023. How urban development affects green development efficiency in China: taking the city cluster of Yangtze river economic belt as an example. *Front. Env. Sci. -Switz.* 11, 1169226.
- Hummera, S., Muhammad, B.K., Seyed, M.M., 2022. The role of green growth, green financing, and eco-friendly technology in achieving environmental quality: evidence from selected Asian economies. *Environ. Sci. Pollut. R.* 29, 57720–57739.
- Ibrahim, R., L., 2022b. Exploring the nexuses of disintegrated energy consumption, structural change, and financial development on environmental sustainability in BRICS: modulating roles of green innovations and regulatory quality. *Sustain. Energy Techn.* 53, 102529.
- Ibrahim, R., L., 2022a. Exploring the nexuses of disintegrated energy consumption, structural change, and financial development on environmental sustainability in BRICS: modulating roles of green innovations and regulatory quality. *Sustain. Energy Techn.* 53, 102529.
- Jiang, P., Zhang, H.W., Li, M.Y., Zhang, Y.H., Gong, X.J., He, D., Liu, L., 2023. Research on the structural optimization of the clean energy industry in the context of dual carbon strategy—A case study of Sichuan Province, China. *Sustainability* 15, 2993.
- Jim, C., 2014. Green-space preservation and allocation for sustainable greening of compact cities. *Cities* 21 (4), 311–320.
- Khan, I., Zakari, A., Dagar, V., Singh, S., 2022a. World energy trilemma and transformative energy developments as determinants of economic growth amid environmental sustainability. *Energ. Econ.* 108, 105884.
- Khan, I., Zakari, A., Zhang, J.J., Dagar, V., Singh, S., 2022b. A study of trilemma energy balance, clean energy transitions, and economic expansion in the midst of

- environmental sustainability: new insights from three trilemma leadership. *Energy* 248, 123619.
- Lau, C.K., Gozgor, G., Mahalik, M.K., 2023. Introducing a new measure of energy transition: green quality of energy mix and its impact on CO₂ emissions. *Energ. Econ.* 122, 106702.
- Li, M.M., Cao, G.K., Li, H.B., Hao, Z.X., Zhang, L., 2023c. How government subsidies affect technology innovation in the context of Industry 4.0: evidence from Chinese new-energy enterprises. *Kybernetes*.
- Li, Z.Z., Ling, J., 2023. The impact of manufacturing intelligence on green development efficiency: a study based on Chinese data. *Sustainability* 15, 7553.
- Li, W., Xu, X.M., Huang, S.Z., Cheng, T., Liu, M.K., Zhang, C., 2023b. Assessment of green technology innovation on energy-environmental efficiency in China under the influence of environmental regulation considering spatial effects. *Sci. Rep.* 13, 20789.
- Li, W.Q., Zhang, F., Pan, L.Y., Li, Z., 2023a. Scenario analysis of carbon emission trajectory on energy system transition model: a case study of Sichuan Province. *Energy Strateg. Rev.* 45, 101015.
- Liang, C.Y., Zhu, L., Zeng, J.B., 2024. Analysis on the impact of media attention on energy enterprises and energy transformation in China. *Energy Rep.* 11, 1820–1833.
- Liu, D., 2021. Application and research of analytic hierarchy process in green GDP development planning of smart city. *J. Urban Plan. Dev.* 147 (1), 04020051.
- Liu, C.M., Tang, C., Liu, Y.D., 2023. Does the transformation of energy structure promote green technological innovation? A quasi-natural experiment based on new energy demonstration city construction. *Geosci. Front.*, 101615.
- Long, R.Y., Li, H.F., Wu, M.F., Li, W.B., 2021. Dynamic evaluation of the green development level of China's coal-resource-based cities using the TOPSIS method. *Resour. Policy* 74, 102415.
- Lu, W.T., Zhang, D.X., 2023. Green development efficiency of urban agglomerations in a developing country: evidence from Beijing-Tianjin-Hebei in China. *Environ. Dev. Sustain.* 25 (7), 6939–6962.
- Lyu, Y.W., Zhang, J.N., Yang, F., Wu, D., 2022. The "local neighborhood" effect of environmental regulation on green innovation efficiency: evidence from China. *Int. J. Env. Res. Pub. He.* 19 (16), 10389.
- Olawumi, T., Chan, D., 2018. A Scientometric review of global research on sustainability and sustainable development. *J. Clean. Prod.* 183, 231–250.
- Saša Stjepanović, S., Tomic, D., Skare, M., 2017. A new approach to measuring green GDP: a cross-country analysis. *Entrep. Sustain.* 4, 574–590.
- Shen, W.F., Liu, Y.F., Liu, X.W., Shi, J.N., Liu, W.B., Liu, C.Y., 2023. The effect of industrial structure upgrading and human capital structure upgrading on green development efficiency—Based on China's resource-based cities. *Sustainability* 15 (5), 4673.
- Shi, B., Yang, H., Wang, J., Zhao, J., 2016. City green economy evaluation: empirical evidence from 15 sub-provincial cities in China. *Sustainability* 8, 551.
- Sun, Y., 2022. Climate and energy governance for the UK low carbon transition. *Carbon Manag* 13, 178–180.
- Tang, Y., Yuan, Y.B., Tian, B.Q., 2023. Assessment of spatio-temporal evolution trends and driving factors of green development in Harbin-Changchun urban agglomeration. *Sci. Rep.* -uk. 13 (1), 16785.
- Tariq, G., Sun, H., Ali, I., Pasha, A.A., Khan, M.S., Rahman, M.M., Mohamed, A., Shah, Q., 2022. Influence of green technology, green energy consumption, energy efficiency, trade, economic development and FDI on climate change in South Asia. *Sci. Rep.* 12, 16376.
- Tian, S., Zhou, X., 2022. Green assessing the impact of digital economy on green development efficiency in the Yangtze River Economic Belt. *Energy Econ.* 112, 106127.
- Tone, K., Tsutsui, M., 2009. Network DEA: a slacks-based measure approach. *Eur. J. Oper. Res.* 197 (1), 243–252.
- Udemba, E.N., Dagar, V., Peng, X., Dagher, L., 2023. Attaining environmental sustainability amidst the interacting forces of natural resource rent and foreign direct investment: is Norway any different? *OPEC Energy Rev.* 12292.
- Wan, Y.Y., 2023. Study on the spatial spillover effect and path mechanism of green finance development on China's energy structure transformation. *J. Clean. Prod.* 415, 137820.
- Wan, Y.Y., Sheng, N., Wei, X.Y., Su, H.Y., 2023. Study on the spatial spillover effect and path mechanism of green finance development on China's energy structure transformation. *J. Clean. Prod.* 415, 137820.
- Wang, X., Sun, C., Wang, S., Zhang, Z., Zou, W., 2018. Going green or going away? A spatial empirical examination of the relationship between environmental regulations, biased technological progress, and green total factor productivity. *Int. J. Environ. Res. Public Health* 15, 1917.
- Wang, F., Wu, M., Wang, J.C., 2023. Can increasing economic complexity improve China's green development efficiency? *Energ. Econ.* 117, 106443.
- Wei, Z.N., Yang, L., Chen, S., Ma, Z.J., Zang, H.X., Fei, Y.D., 2022. A multi-stage planning model for transitioning to low-carbon integrated electric power and natural gas systems. *Energy* 254, 124361.
- Wojtkowska-Lodej, G., Jakubów, E., 2022. The role of clean generation technologies in the energy transformation in Poland. *Energies* 15, 4863.
- Xu, Y., Ge, W.F., Liu, G.L., Su, X.F., Zhu, J.N., Yang, C.Y., Yang, X.D., Ran, Q.Y., 2023. The impact of local government competition and green technology innovation on economic low-carbon transition: new insights from China. *Environ. Sci. Pollut. Res.* 30, 23714–23735.
- Xu, S., Zhou, Y., 2023. OFDI, industrial structure upgrading and green development—spatial effect based on China's evidence. *Sustainability* 15 (3), 2810.
- Yang, Z.B., Wang, X.J., 2022. Research on low-carbon capability evaluation model of city regional integrated energy system under energy market environment. *Processes* 10, 1906.
- Yang, H., D., Zhang, X., L., Liu, B.Y., Huang, Y.L., 2023. Evaluation of green low-carbon innovation development efficiency: an improved two-stage non-cooperative DEA mode. *J. Clean. Prod.* 400, 136662.
- Yu, D.S., Li, X.P., Yu, J.J., Li, H., 2021. The impact of the spatial agglomeration of foreign direct investment on green total factor productivity of Chinese cities. *J. Environ. Manag.* 290, 112666.
- Yu, S.W., Zheng, Y.L., Hu, X., Shu, K.S., 2022. Spatial impacts of biomass resource endowment on provincial green development efficiency. *Renew. Energ.* 189, 651–662.
- Yue, S., Yang, Y., Hu, Y., 2016. Does foreign direct investment affect green growth? Evidence from China's experience. *Sustainability* 8, 158.
- Zakari, A., Khan, I.A., Tan, D.J., Alvarado, R., Dagar, V., 2022. Energy efficiency and sustainable development goals (SDGs). *Energy* 239, 122365.
- Zhan, L., Guo, P., Pan, G.Q., 2023. The effect of mandatory environmental regulation on green development efficiency: evidence from China. *Environ. Sci. Pollut. R.* 30 (4), 9782–9792.
- Zhang, Y.S., Zhou, R.M., Chen, N.X., 2023. An empirical study on coordinated development of energy consumption structure and green total factor productivity under spatial interaction. *Econ. Res-Ekon. Istraz.* 36 (1), 1226–1246.
- Zhao, C.Y., Dong, K.Y., 2023. Can smart transportation enhance green development efficiency? *Econ. Change Restruct.* 56 (2), 825–857.
- Zhao, M., Sun, T., Feng, Q., 2021. Capital allocation efficiency, technological innovation and vehicle carbon emissions: evidence from a panel threshold model of Chinese new energy vehicles enterprises. *Sci. Total Environ.* 784, 147104.
- Zhou, M.C., Fan, R., 2023. The impact of green capital structure on enterprise development and the regulation of technological innovation under carbon peaking. *Sustainability* 15, 5743.
- Zhu, B., Zhang, M., Zhou, Y., Wang, P., Sheng, J., He, K., Xie, R., 2019. Exploring the effect of industrial structure adjustment on interprovincial green development efficiency in China: a novel integrated approach. *Energ. Policy* 134, 110946.
- Zou, T., Guo, P.B., Wu, Q.L., 2023. Applying an entropy-weighted TOPSIS method to evaluate energy green consumption revolution progressing of China. *Environ. Sci. Pollut. R.* 30 (14), 42267–42281.